Model Setup

Without loss of generality, let the predictor variables $\{x_1, x_2, x_3, x_4\}$ be standardized with zero means, unit variance and correlations ρ. We assume that the predictor variables come from a multivariate normal distribution, that is:

$$(x_{1i}, x_{2i}, x_{3i}, x_{4i}) \sim \mathcal{N}(0, \Sigma) \quad \text{where} \quad \Sigma = \begin{pmatrix} 1 & \rho & \rho & \rho \\ & 1 & \rho & \rho \\ & & 1 & \rho \\ & & & 1 \end{pmatrix} \tag{16}$$

Let y denote the outcome variable related to the predictor variables $\{x_1, x_2, x_3, x_4\}$ in accordance with the following model:

$$\begin{aligned} y_i =& \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \beta_4 x_{4i} + & (17) \\ & \beta_5 x_{1i}x_{2i} + \beta_6 x_{1i}x_{3i} + \beta_7 x_{1i}x_{4i} + \beta_8 x_{2i}x_{3i} + \beta_9 x_{2i}x_{4i} + \beta_{10} x_{3i}x_{4i} + & (18) \\ & \beta_{11} x_{1i}^2 + \beta_{12} x_{2i}^2 + \beta_{13} x_{3i}^2 + \beta_{14} x_{4i}^2 + \varepsilon_i & (19) \\ \text{with} \quad \varepsilon_i \sim& \mathcal{N}(0, \sigma_\varepsilon^2) & (20) \end{aligned}$$

In matrix notation, we can write:

$$y = X\beta + \varepsilon, \quad \varepsilon \sim \mathcal{N}(\mathbf{0}, \sigma_\varepsilon^2 I) \tag{21}$$

where X is the matrix of predictor values and β is the vector of regression weights.

We model effects for the (linear) predictors and an identical number of interactions between predictors. Thus, the regression weights for two of the possible interactions (i.e., $x_{1i}x_{3i}$ and $x_{2i}x_{4i}$) need to be 0 ($\{\beta_6, \beta_9\} = 0$). At the moment we simulate without polynomials, i.e., $\{\beta_{11}, \beta_{12}, \beta_{13}, \beta_{14}\} = 0$.

Covariance matrix of predictors

Using the results from of Bohrnstedt and Goldberger (1969) and assuming multivariate normality of the predictor variables $\{x_1, x_2, x_3, x_4\}$, it can be shown that the covariance matrix of the predictors is given by

$$\Sigma_X = \begin{pmatrix} 1 & \rho & \rho & \rho & 0 & 0 & 0 & 0 & 0 & 0 \\ & 1 & \rho & \rho & 0 & 0 & 0 & 0 & 0 & 0 \\ & & 1 & \rho & 0 & 0 & 0 & 0 & 0 & 0 \\ & & & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ & & & & 1+\rho^2 & \rho+\rho^2 & \rho+\rho^2 & \rho+\rho^2 & \rho+\rho^2 & 2\rho^2 \\ & & & & & 1+\rho^2 & \rho+\rho^2 & \rho+\rho^2 & 2\rho^2 & \rho+\rho^2 \\ & & & & & & 1+\rho^2 & 2\rho^2 & \rho+\rho^2 & \rho+\rho^2 \\ & & & & & & & 1+\rho^2 & \rho+\rho^2 & \rho+\rho^2 \\ & & & & & & & & 1+\rho^2 & \rho+\rho^2 \\ & & & & & & & & & 1+\rho^2 \end{pmatrix} \tag{22}$$

It should be noted that this matrix has a block diagonal structure:

$$\Sigma_X = \begin{pmatrix} V_1 & \mathbf{0} \\ \mathbf{0} & V2 \end{pmatrix} \tag{23}$$

Therefore, linear effects and interaction effects as simulated within a linear regression framework are independent from each other.

The correlation matrix of the predictors is given by:

$$R_X = \begin{pmatrix} 1 & \rho & \rho & \rho & 0 & 0 & 0 & 0 & 0 & 0 \\ & 1 & \rho & \rho & 0 & 0 & 0 & 0 & 0 & 0 \\ & & 1 & \rho & 0 & 0 & 0 & 0 & 0 & 0 \\ & & & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ & & & & 1 & \frac{\rho+\rho^2}{1+\rho^2} & \frac{\rho+\rho^2}{1+\rho^2} & \frac{\rho+\rho^2}{1+\rho^2} & \frac{\rho+\rho^2}{1+\rho^2} & \frac{2\rho^2}{1+\rho^2} \\ & & & & & 1 & \frac{\rho+\rho^2}{1+\rho^2} & \frac{\rho+\rho^2}{1+\rho^2} & \frac{2\rho^2}{1+\rho^2} & \frac{\rho+\rho^2}{1+\rho^2} \\ & & & & & & 1 & \frac{2\rho^2}{1+\rho^2} & \frac{\rho+\rho^2}{1+\rho^2} & \frac{\rho+\rho^2}{1+\rho^2} \\ & & & & & & & 1 & \frac{\rho+\rho^2}{1+\rho^2} & \frac{\rho+\rho^2}{1+\rho^2} \\ & & & & & & & & 1 & \frac{\rho+\rho^2}{1+\rho^2} \\ & & & & & & & & & 1 \end{pmatrix} \tag{24}$$

Note: $\sigma^2_{x_1x_2} = 1 + \rho^2$ and $\sigma^2_{x_1x_3} = 1 + \rho^2$, thus $\sigma_{x_1x_2}\sigma_{x_1x_3} = 1 + \rho^2$ instead of

$\sigma_{x_1x_2}\sigma_{x_1x_3} = \sqrt{1+\rho^2}\sqrt{1+\rho^2}$.

Starting from the results of Bohrnstedt and Goldberger (1969) and assuming bivariate normality of x and y, the variance of the product of x and y is.

$$\begin{aligned}
V(xy) =& E^2(x)V(y) + E^2(y)V(x) + 2E(x)E(y)C(x,y) + V(x)V(y) + C^2(x,y) \\
\sigma^2_{xy} =& (\mu_x\sigma_y)^2 + (\mu_y\sigma_x)^2 + 2\mu_x\mu_y\rho\sigma_x\sigma_y + \sigma_x^2\sigma_y^2 + (\rho\sigma_x\sigma_y)^2 \\
=& (\mu_x\sigma_y)^2 + (\mu_y\sigma_x)^2 + 2\mu_x\mu_y\rho\sigma_x\sigma_y + (\sigma_x\sigma_y)^2 + \rho^2(\sigma_x\sigma_y)^2 \\
=& (\mu_x\sigma_y)^2 + (\mu_y\sigma_x)^2 + 2\mu_x\mu_y\rho\sigma_x\sigma_y + (1+\rho^2)(\sigma_x\sigma_y)^2
\end{aligned}$$

For standardized variables with zero means ($\mu_x = \mu_y = 0$)and unit variance ($\sigma_x = \sigma_y = 1$):

$$\begin{aligned}
\sigma^2_{xy} =& (\mu_x\sigma_y)^2 + (\mu_y\sigma_x)^2 + 2\mu_x\mu_y\rho\sigma_x\sigma_y + (1+\rho^2)(\sigma_x\sigma_y)^2 \\
=& (0\cdot 1)^2 + (0\cdot 1)^2 + 2\cdot 0\cdot 0\cdot \rho\cdot 1\cdot 1 + (1+\rho^2)(1\cdot 1)^2 \\
=& (1+\rho^2)
\end{aligned}$$

The covariance of products of random variables from a multivariate normal distribution is denoted by:

$$\begin{aligned}
C(xy,uv) =& E(x)E(u)C(y,v) + E(x)E(v)C(y,v) + E(y)E(u)C(x,v) + E(y)E(v)C(x,u) + \\
& C(x,u)C(y,v) + C(x,v)C(y,u) \\
=& \mu_x\mu_u\rho_{yv}\sigma_y\sigma_v + \mu_x\mu_v\rho_{yv}\sigma_y\sigma_v + \mu_y\mu_u\rho_{xv}\sigma_x\sigma_v + \mu_y\mu_v\rho_{xu}\sigma_x\sigma_u + \\
& \rho_{xu}\sigma_x\sigma_u\cdot\rho_{yv}\sigma_y\sigma_v + \rho_{xv}\sigma_x\sigma_v\cdot\rho_{yu}\sigma_y\sigma_u
\end{aligned}$$

For standardized variables with zero means ($\mu_x = \mu_y = \mu_u = \mu_v = 0$) and unit variance

$(\sigma_x = \sigma_y = \sigma_u = \sigma_v = 1)$:

$$\begin{aligned}
C(xy, uv) =& \mu_x\mu_u\rho_{yv}\sigma_y\sigma_v + \mu_x\mu_v\rho_{yv}\sigma_y\sigma_v + \mu_y\mu_u\rho_{xv}\sigma_x\sigma_v + \mu_y\mu_v\rho_{xu}\sigma_x\sigma_u + \\
& \rho_{xu}\sigma_x\sigma_u \cdot \rho_{yv}\sigma_y\sigma_v + \rho_{xv}\sigma_x\sigma_v \cdot \rho_{yu}\sigma_y\sigma_u \\
=& 0 \cdot 0 \cdot \rho_{yv} \cdot 1 \cdot 1 + 0 \cdot 0 \cdot \rho_{yv} \cdot 1 \cdot 1 + 0 \cdot 0 \cdot \rho_{xv} \cdot 1 \cdot 1 + 0 \cdot 0 \cdot \rho_{xu} \cdot 1 \cdot 1 + \\
& \rho_{xu} \cdot 1 \cdot 1 \cdot \rho_{yv} \cdot 1 \cdot 1 + \rho_{xv} \cdot 1 \cdot 1 \cdot \rho_{yu} \cdot 1 \cdot 1 \\
=& \rho_{xu} \cdot \rho_{yv} + \rho_{xv} \cdot \rho_{yu}
\end{aligned}$$

For the present simulation model, we need to differentiate two cases which are represented by the following two examples:

- The products share one of the variables.

$$\begin{aligned}
C(xy, uv) =& \rho_{xu} \cdot \rho_{yv} + \rho_{xv} \cdot \rho_{yu} \\
C(x_1x_2, x_1x_3) =& \rho_{x_1x_1} \cdot \rho_{x_2x_3} + \rho_{x_1x_3} \cdot \rho_{x_2x_1} \\
C(x_1x_2, x_1x_3) =& 1 \cdot \rho + \rho \cdot \rho \\
C(x_1x_2, x_1x_3) =& \rho + \rho^2
\end{aligned}$$

- The products do not share any variable.

$$\begin{aligned}
C(xy, uv) =& \rho_{xu} \cdot \rho_{yv} + \rho_{xv} \cdot \rho_{yu} \\
C(x_1x_2, x_3x_4) =& \rho_{x_1x_3} \cdot \rho_{x_2x_4} + \rho_{x_1x_4} \cdot \rho_{x_2x_3} \\
C(x_1x_2, x_1x_3) =& \rho \cdot \rho + \rho \cdot \rho \\
C(x_1x_2, x_1x_3) =& 2\rho^2
\end{aligned}$$

Piecewise-linear DGP: second segment variable variance

We sample the original predictor variable from $X \sim \mathcal{N}(\mu, \sigma^2)$ with $\mu = 0$ and $\sigma^2 = 1$. To arrive at the predictor variable for the second segment, we transform X into X^* with

$$x_i^* = (x_i - \psi) \cdot dummy_{x_i} \text{ with } dummy_{x_{pi}} = \begin{cases} 0, & \text{if } x_{pi} \leq \psi \\ 1, & \text{if } x_{pi} > \psi \end{cases}$$

For $\psi = 0$, the given transformation is equivalent to applying the rectified linear unit (ReLU) activation function to the original predictor (or using a rectified Gaussian distribution to sample from).

$$\begin{aligned} x_i^* &= (x_i - 0) \cdot dummy_{x_i} \\ &= max(0, x_i) \end{aligned}$$

Without loss of generality, we can use the probability density function (PDF) of the standard normal distribution $\phi(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right)$ and the corresponding cumulative distribution function (CDF) $\Phi(x) = \int_{-\infty}^{x} \phi(t)\, \mathrm{d}t$ to calculate the variance of X^*.

The variance of a variable is given by

$$\mathrm{Var}(X^*) \; = \; E\left[(X^*)^2\right] \; - \; \left(E[X^*]\right)^2$$

The expected value of a ReLU transformed variable X^* is given by

$$E[X^*] \; = \; \mu\left[1 - \Phi(-\frac{\mu}{\sigma})\right] \; + \; \sigma\,\phi(-\frac{\mu}{\sigma})$$

and its second moment with

$$E\left[(X^*)^2\right] \; = \; (\mu^2 + \sigma^2)\left[1 - \Phi(-\frac{\mu}{\sigma})\right] \; + \; \mu\,\sigma\,\phi(-\frac{\mu}{\sigma})$$

with $-\frac{\mu}{\sigma}$ essentially giving the z-score of the breakpoint (Beauchamp, 2018).

By substituting the respective expressions with the simulated true values of the original variable X, we obtain the variance of the variable X^*.